

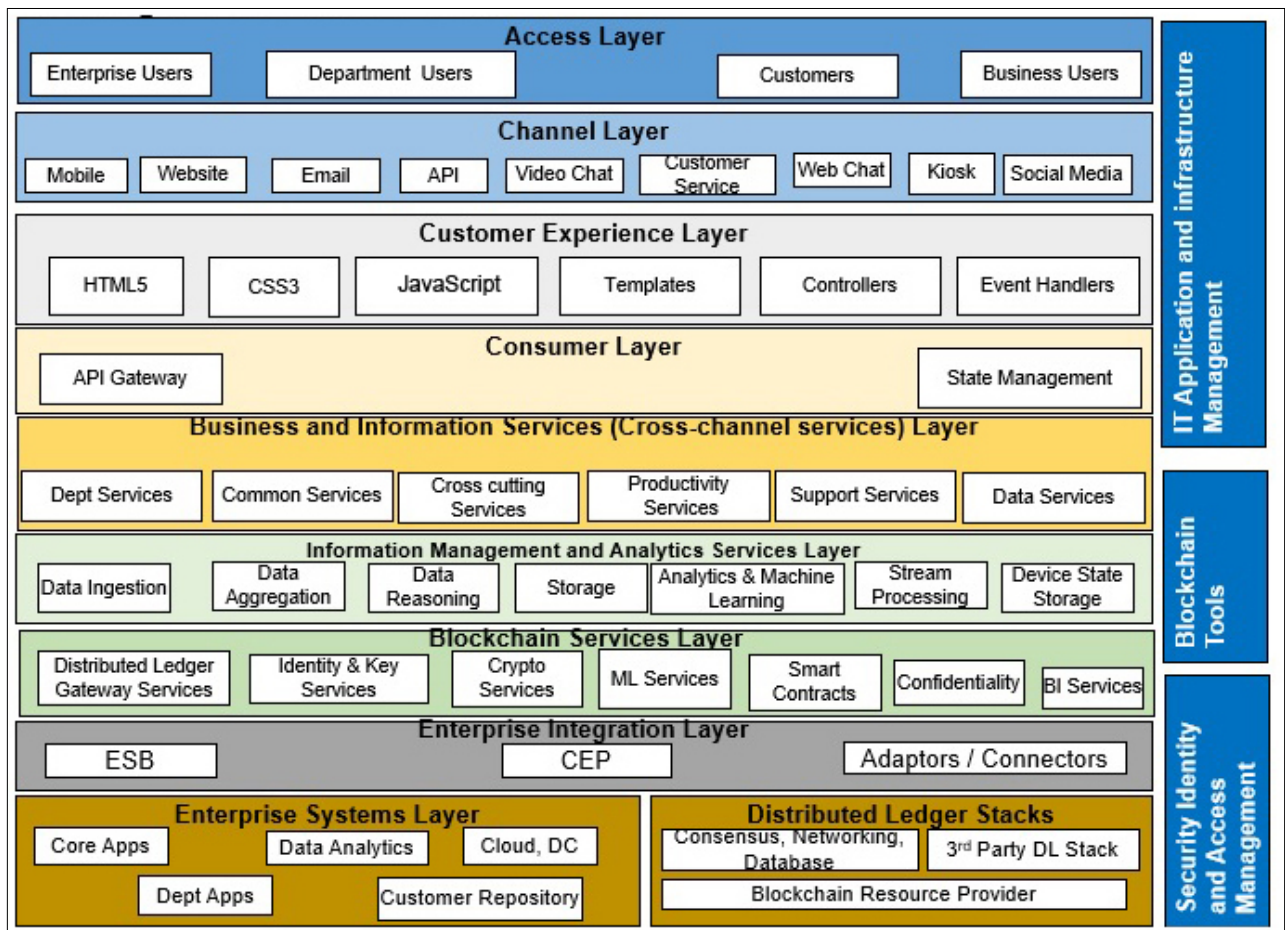


Figure 2: Blockchain logical reference architecture

Real world use cases of blockchain technology

Government

In government, linking the data between various departments with blockchain ensures that the data is realised in real-time. Blockchain technology could improve transparency and check corruption in governments worldwide.

- **Registration:** By securing a unique and non-corruptible record on a blockchain and validating changes to the status of that record across owners, a reliable property record can be created, whether for a piece of land that heretofore had no owner or as a link between stovepiped systems. A decentralised, standardised system for land registration records could reduce the number of intermediaries required, increase trust in the identity of transacting parties, increase process efficiencies, and decrease time and cost to process.
- **Election:** Citizens can cast votes the same way they initiate other secure transactions and validate that their votes were cast—or even verify the election results. Solutions are being worked on to blend secure digital identity management, anonymous vote-casting, individualised ballot processes, and ballot casting confirmation verifiable by the voter. Potential cost savings can be achieved through blockchain-enabled voting.
- **Finance:** Online identity has always been a time-consuming and costly process. Financial services like loans, mortgages, and insurance have always required a higher level of security, checking, authentication and authorisation. This meant that checking official government identity documents was a required step. One of the benefits of the blockchain is that it has the potential to cut out the intermediaries and provide every organisation/department in the network access to the same source of the right information. Blockchain technologies make tracking and managing digital identities secure and efficient, resulting in reduced fraud. Users are able to choose how they identify themselves and with whom their identity is shared.
- **Town planning:** Having property records on the blockchain makes it possible that prospective buyers shall